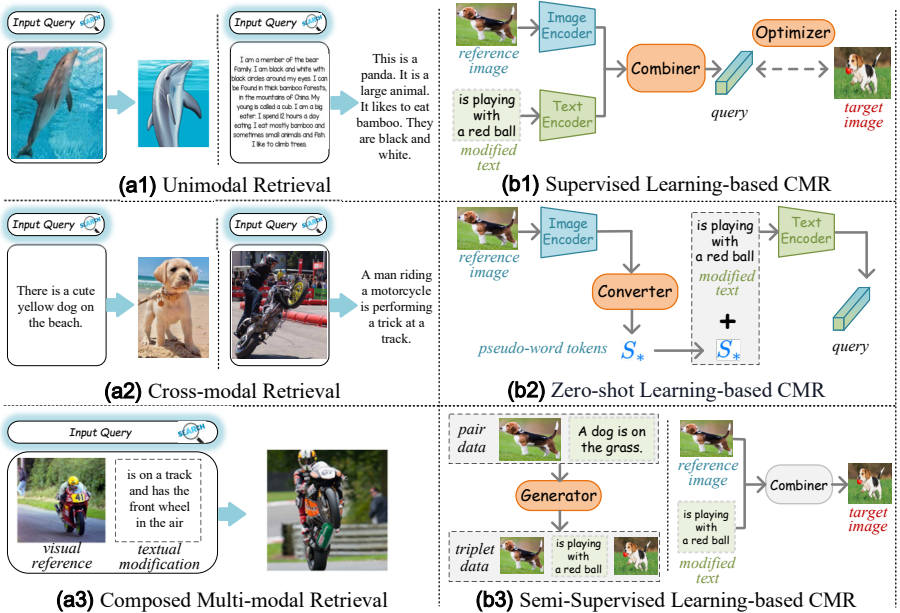
Input Query
I am a member of the bear Family. I am black and white with black circles around my eyes. I can be Found in thick bamboo Forests, in the mountains of China. My young is called a cub. I am a big eater. I spend 12 hours a day eating. I eat mostly bamboo and sometimes small animals and Fish. I like to climb trees.
This is a panda. It is a large animal. It likes to eat bamboo. They are black and white.
(a1) Unimodal Retrieval
There is a cute yellow dog on the beach.
A man riding a motorcycle is performing a trick at a track.
(a2) Cross-modal Retrieval
is on a track and has the front wheel in the air
visual reference
textual modification
(a3) Composed Multi-modal Retrieval
reference image
Image Encoder
is playing with a red ball
modified text
Text Encoder
Combiner
Optimizer
query
target image
(b1) Supervised Learning-based CMR
reference image
Image Encoder
Converter
pseudo-word tokens
S_*
is playing with a red ball
modified text
Text Encoder
query
(b2) Zero-shot Learning-based CMR
pair data
A dog is on the grass.
Generator
triplet data
is playing with a red ball
reference image
is playing with a red ball
modified text
Combiner
target image
(b3) Semi-Supervised Learning-based CMR
Composed Multi-modal Retrieval
Composed Fashion Image Retrieval
Similar one but long
Composed Natural Image Retrieval
Replace bread with boiled meat
Composed Video Retrieval
Show it at night
Composed Remote Sensing Image Retrieval
With four similar ones
Composed Person Retrieval
Wearing a silver necklace, and carrying a green bag
Composed Sketch-based Image Retrieval
with brown leather and silver armrests
Interactive/Conversational Retrieval
(c) Applications